

The Gap Between Mandate and Execution: An Evaluation of the Inflation Target in Costa Rica

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Abstract

This paper addresses recent concerns about a possible contractionary bias in the policy execution of the Central Bank of Costa Rica (BCCR). I evaluate this hypothesis using three empirical approaches: a VAR estimation of long-run expectations following [Demertzis et al. \(2008\)](#), a UCSV estimation of the long-run inflationary trend based on [Chan et al. \(2018\)](#), and a structural estimation of policy rule asymmetries following [Surico \(2007\)](#). The results show that expectations stabilize towards an anchor between 2.46-2.58%, lower than the BCCR's 3% mandate. The estimated inflationary trend shows systematic deviations from the target, with its 2023-2025 varying between 1.98-2.35%. Finally, I find statistically significant evidence of an asymmetric policy rule, characterized by a more aggressive reaction to positive inflation deviations relative to negative ones.

1 Introduction

Monetary policy and its implementation play a pivotal role in the maximization of aggregate welfare. In this regard, inflation targeting has been a widespread success since its pioneering implementation in the 1990s, introducing a nominal anchor by which the public can clearly assess the central bank's policy against a clear metric. In the Costa Rican context, the Central Bank of Costa Rica (BCCR) officially adopted this framework in 2018, but its *de facto* implementation dates back to 2005. The BCCR's Board of Directors defined the explicit inflation target at 3%, with a tolerance range of ± 1 percentage point, within a horizon of 24 months. ([Central Bank of Costa Rica, 2025](#)).

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Recent public commentary (e.g. [Martínez, 2025](#); [Ramírez, 2024](#); [Zúñiga, 2024](#)) has criticized the BCCR's policy implementation, arguing that it has been too slow to lower its policy rate relative to the rapid drop in inflation. More specifically, the BCCR is accused of having an implicitly contractionary bias, with its response being aggressive when inflation is above the target, but timid when it is below the target. This argument is supported by recent observations of CPI inflation: only 1 of 35 monthly observations between 2023 and November 2025 has fallen within the tolerance range, as depicted in [Figure 1](#). Considering the period from the official adoption of the target in January 2018 to November 2025, only 25 out of 95 monthly observations (26 %) have remained within the tolerance range. Additionally, the BCCR has continuously deferred the return of inflation to the target range, as shown in their Monetary Policy Reports:

“...return to the tolerance range in the Q2 2024.”

- ([Central Bank of Costa Rica](#), October 2023 Monetary Policy Report)

“...return to the tolerance range in Q3 2025.”

- ([Central Bank of Costa Rica](#), October 2024 Monetary Policy Report)

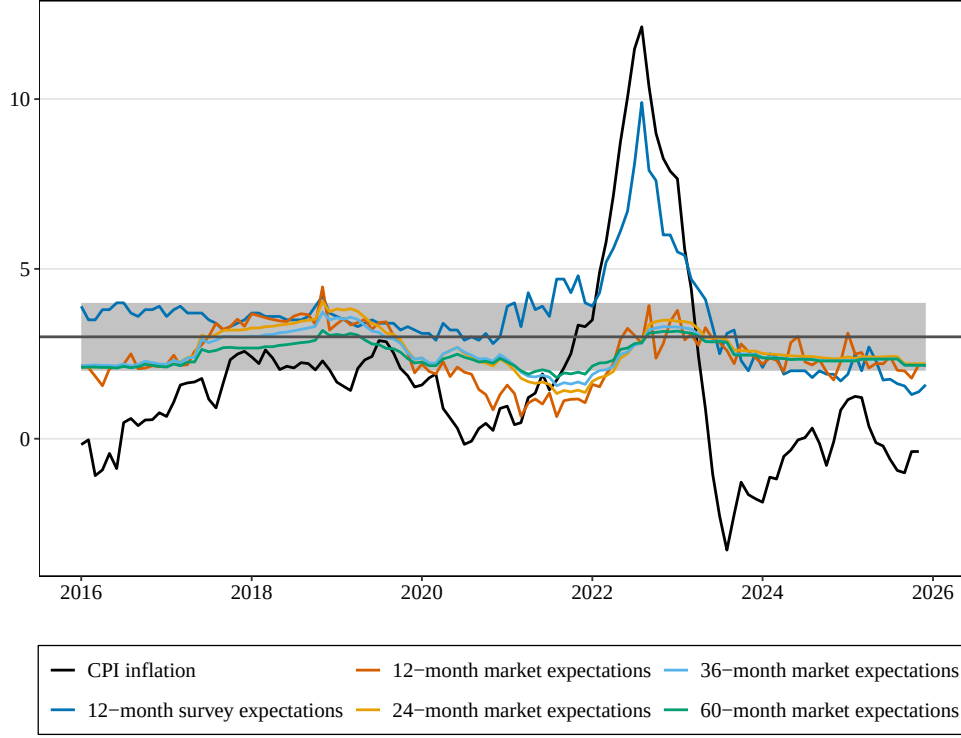
“Return to the tolerance range in Q2 2027 for general inflation and in Q3 2026 for core inflation.”

- ([Central Bank of Costa Rica](#), October 2025 Monetary Policy Report)

To measure the extent of this bias, I implement several empirical methodologies. First, I use a VAR approach introduced by [Demertzis et al. \(2008\)](#) to estimate the long-run expectation anchorage in Costa Rica. Second, I implement an Unobserved Components with Stochastic Volatility (UCSV) model introduced by [Chan et al. \(2018\)](#), which relies on long-run inflation expectations and actual inflation to recover a long-run inflationary trend. Finally, drawing on the structural framework of [Surico \(2007\)](#), I test for nonlinearities in the BCCR's reaction function, assessing whether the authority weighs inflation overshooting differently than undershooting.

While inflation expectations are strongly anchored, they have converged to an implicit level between 2.46% and 2.58%, significantly below the official 3% target. The estimated long-run inflationary trend exhibits persistent downward deviations from the inflation target, with its average between 2023 and 2025 varying between 1.98% and 2.35% for different specifications. Finally, the structural analysis shows that, given its explicit target, the BCCR has displayed a statistically significant aversion to inflation overshooting relative to undershooting.

Figure 1. Costa Rica. Inflation, expectations and target range. 2016-2025



Notes: The figure shows the evolution of CPI inflation and various measures of inflation expectations. The gray horizontal line marks the 3% target value, while the shaded line corresponds to the ± 1 % tolerance range. Market-based expectations are recovered from the negotiations of domestic debt securities in national currency carried out in the local market and published by the BCCR.

The rest of the article is structured as follows. Section 2 introduces the methodological frameworks to be used for the three exercises. Section 3 presents and explains the key results of each exercise. Section 4 presents some concluding remarks.

2 Methodology

2.1 Estimation of the Long-run Expectation Anchor

The estimation of the inflation anchor builds on [Bomfim & Rudebusch \(2000\)](#), who assume that long-run inflation expectations (z_t) are composed as a weighted average of a nominal anchor z^* and past inflation from the last period π_{t-1} :

$$z_t = \chi z^* + (1 - \chi)\pi_{t-1}, \quad (1)$$

where $\chi \in [0, 1]$ indexes the degree of anchorness of inflation expectations. [Demertzis et al. \(2008\)](#) propose an endogenous estimation method for the nominal anchor and the

degree of anchorness, which consists of estimating the following VAR(1) model:

$$\pi_t = \alpha_0 + \alpha_1 \pi_{t-1} + \alpha_2 z_{t-1} + \epsilon_{1t} \quad (2)$$

$$z_t = \beta_0 + \beta_1 \pi_{t-1} + \beta_2 z_{t-1} + \epsilon_{2t}. \quad (3)$$

By considering the long-run equilibrium equations generated by the system, particularly the equation of inflation expectations, we have

$$z_t = \frac{\beta_0}{1 - \beta_2} + \frac{\beta_1}{1 - \beta_2} \pi_t. \quad (4)$$

Finally, z^* and χ can be recovered from the VAR parameters by matching coefficients of equations (1) and (4):

$$\chi = 1 - \frac{\beta_1}{1 - \beta_2}, \quad z^* = \frac{\beta_0}{(1 - \beta_2)\chi}. \quad (5)$$

2.2 Estimation of the Long-run Inflationary Trend

The long-run inflationary trend π_t^* is defined following [Beveridge & Nelson \(1981\)](#). Formally, we have

$$\pi_t^* \equiv \lim_{h \rightarrow \infty} \mathbb{E}_t[\pi_{t+h} | \Omega_t], \quad (6)$$

where $\mathbb{E}_t$ denotes the expectation operator conditional to the information set at period t Ω_t . This definition allows inflation to be broken down into a permanent component and a transitory component $\tilde{\pi}_t$, that is, $\pi_t = \pi_t^* + \tilde{\pi}_t$.

Following [Chan et al. \(2018\)](#), I assume the dynamic of inflation, expectations and the long-run inflationary trend is described by the following equations:

$$\pi_t - \pi_t^* = b_t(\pi_{t-1} - \pi_{t-1}^*) + v_t, \quad (7)$$

$$z_t = d_{0t} + d_{1t}\pi_t^* + \varepsilon_{z,t} + \psi\varepsilon_{z,t-1}, \quad \varepsilon_{z,t} \sim N(0, \sigma_z^2) \quad (8)$$

$$\pi_t^* = \pi_{t-1}^* + n_t, \quad (9)$$

$$b_t = b_{t-1} + \varepsilon_{b,t}, \quad \varepsilon_{b,t} \sim TN(0, \sigma_b^2) \quad (10)$$

$$d_{it} - \mu_{di} = \rho_{di}(d_{i,t-1} - \mu_{di}) + \varepsilon_{di,t}, \quad \varepsilon_{di,t} \sim N(0, \sigma_{di}^2), \quad i = 0, 1 \quad (11)$$

$$v_t = \lambda_{v,t}^{0.5} \varepsilon_{v,t}, \quad \varepsilon_{v,t} \sim N(0, 1), \quad (12)$$

$$n_t = \lambda_{n,t}^{0.5} \varepsilon_{n,t}, \quad \varepsilon_{n,t} \sim N(0, 1), \quad (13)$$

$$\log \lambda_{i,t} = \log \lambda_{i,t-1} + v_{i,t}, \quad v_{i,t} \sim N(0, \phi_i), \quad i = v, n. \quad (14)$$

Equation (7) describes the evolution of the cyclical component of inflation, where b_t measures the persistence of cyclical deviations, and its variations allow capturing temporary differences in the response of inflation to transitory shocks. Equation (8) relates long-term inflation expectations to the underlying trend. The linear specification with moving-average errors and time varying coefficients (d_{0t} and d_{1t}) allows for possible temporary deviations in the agents' rational expectations, as those documented by [Coibion & Gorodnichenko \(2015\)](#) and [Fisher et al. \(2025\)](#). Equation (9) models the inflationary trend as a random walk process. Equations (10) and (11) describe the temporal evolution of the coefficients, where TN denotes a truncated normal distribution, which ensures stationarity of the inflation gap $\pi_t - \pi_t^*$. Finally, equations (12), (13), and (14) model the stochastic volatility of the permanent and transitory components of inflation. The main result to be estimated is the posterior mean of the inflation trend. An MCMC algorithm is used to estimate the latent variables and unknown parameters of the model. The prior distributions are kept as in the original paper.

2.3 Testing for asymmetries in the BCCR's response

[Surico \(2007\)](#) introduces a monetary policy framework in which a central bank is allowed to assign different weights on positive and negative deviations of inflation and output from their respective targets. Particularly the central bank has a linex loss function given by

$$L_t = \frac{e^{\alpha(\pi_t - \pi^T)} - \alpha(\pi_t - \pi^T) - 1}{\alpha^2} + \lambda \left(\frac{e^{\gamma x_t} - \gamma y_t - 1}{\gamma^2} \right) + \frac{\mu}{2} (i_t - i_t^*)^2, \quad (15)$$

where π^T is the inflation target, x_t is the output gap, and i^* is the natural interest rate. This function is chosen as it allows for asymmetries in the policy response. Parameters $\lambda, \mu > 0$ reflect the central bank's aversion of output and interest rate fluctuations, while α, γ capture the degree of asymmetry in the central bank's preferences¹. The intertemporal problem of the central bank yields an Euler equation of the form

$$i_t = d_0 + \mathbb{E}_{t-1} \left[d_1 \left(\frac{e^{\alpha(\pi - \pi^T)} - 1}{\alpha} \right) + d_2 \left(\frac{e^{\gamma x_t} - 1}{\gamma} \right) \right], \quad (16)$$

where d_0, d_1, d_2 are functions of the structural parameters. A structural estimation of 16 is plausible only if the data reveals non-linearity in the central bank's response. To test for this, a first-order Taylor approximation of equation (16), with a lagged interest rate term

¹ See [Surico \(2007\)](#) for the intuition behind the linex function and derivations of the policy rule.

for interest smoothing, is estimated and tested for non-linearity:

$$i_t = c_0 + c_1\pi_t + c_2y_t + c_3\pi_t^2 + c_4y_t^2 + v_t, \quad (17)$$

where $c_0, \dots, c_4$ are functions of structural parameters. The restriction $\alpha = \gamma = 0$ corresponds to $c_3 = c_4 = 0$. In this form, the null hypothesis $H_0 : c_3 = c_4 = 0$ can be tested with a standard Wald test.

Equation (17) is estimated by the Generalized Method of Moments (GMM), using $Z_t = \{i_{t-1}, i_{t-2}, \pi_{t-1}, \pi_{t-2}, x_{t-1}, x_{t-2}, \pi_{t-1}^2, \pi_{t-2}^2, x_{t-1}^2, x_{t-2}^2\}$ as the instrument set. Upon rejection of the null hypothesis $H_0 : c_3 = c_4 = 0$, parameters $d_0, d_1, d_2, \alpha, \gamma$ of equation (16) are estimated via non-linear GMM, maintaining Z_t as the instrument set.

2.4 Data and Sources

I employ data on a quarterly frequency for all exercises. Inflation is calculated as a quarterly annualized rate given by $\pi_t = 400 \cdot \log\left(\frac{p_t}{p_{t-1}}\right)$, where p_t is measured as the Consumer Price Index (CPI). In the first two exercises, I include specifications in which p_t is taken as the average of core price indicators published by the BCCR². Inflation expectations are taken as the 60-month market expectations. A moving-average filtered version of this indicator was included in exercise two with the aim of removing additional volatility that these expectations may include due to their construction (see Segura-Rodríguez, 2019). Finally, output gap was constructed as log-deviations of GDP from its trend, calculated using a one-sided HP filter.

3 Results

Table 1 summarizes the results of the first exercise³. In line with the results from Segura-Rodríguez (2019), the implicit anchor varies between 2.46 and 2.58, with the official target being outside the 95% confidence interval for both specifications, but still within the tolerance range. The degree of anchorness varies between 0.82 and 0.91, indicating that economic agents place an important weight to this implied anchor rather than recent inflation history. Overall, the results suggest that, while the BCCR has successfully anchored expectations, it has done so at a level significantly below its explicit target. This

²For exercise three, Sargan–Hansen J -test's null hypothesis was rejected when using core inflation, suggesting that some instruments of Z_t are invalid for this particular specification. Therefore, this specification was discarded.

³See Appendix A for the VAR estimation results.

phenomenon seems to be more evident in CPI inflation than in core inflation, with the latter having a higher implied target and a lower degree of anchorness.

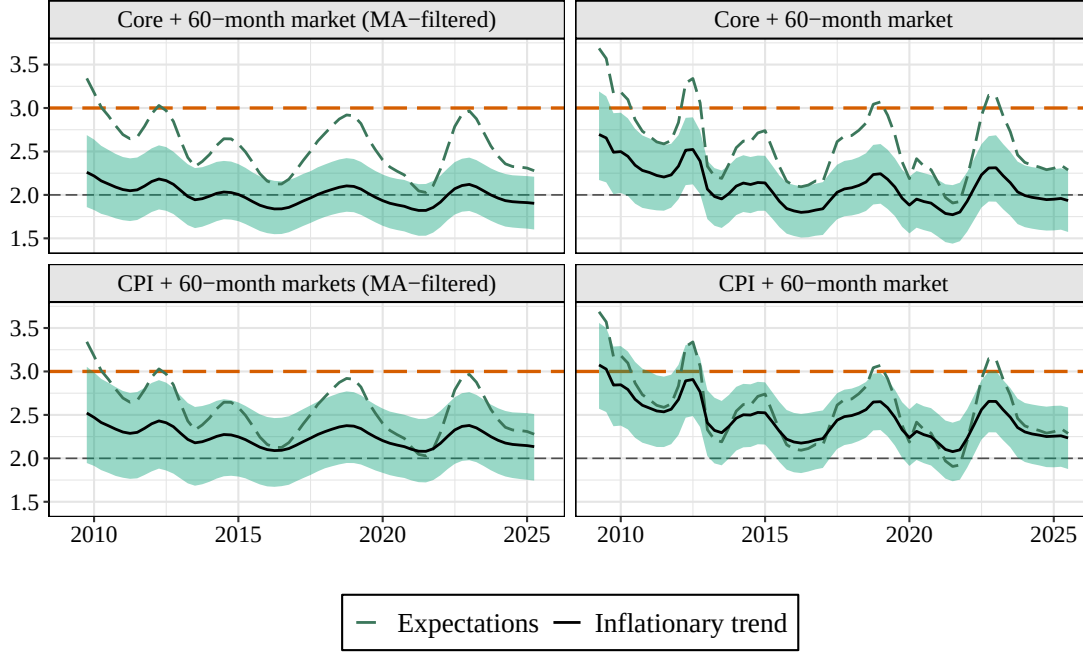
Table 1. Estimates of Anchorness Parameters

Parameter	Estimate	Std. Error	95% Conf. Interval
<i>CPI Inflation</i>			
Anchorness (χ)	0.9098	0.0360	[0.8392, 0.9804]
Implicit anchor (z^*)	2.4594	0.1339	[2.1969, 2.7218]
<i>Core Inflation</i>			
Anchorness (χ)	0.8176	0.0494	[0.7206, 0.9145]
Implicit anchor (z^*)	2.5808	0.1105	[2.3643, 2.7973]

Notes: The table reports estimates of the credibility parameter χ and the implicit anchor z^* , derived from the VAR model following equation (5). Standard errors are calculated using the Delta method based on the covariance matrix of the reduced-form VAR estimates. The 95% confidence intervals are constructed using the critical values from the normal distribution.

Figure 2 shows the estimated trend for all specifications. The inflationary trend has consistently been below the inflation target, and its average between 2023 and 2025 ranges from 1.98% (with core inflation and smoothed expectations) to 2.35% (with CPI inflation and unsmoothed expectations). For all specifications except CPI and unsmoothed expectations, the 84th percentile of the posterior distribution has been below the target value of 3%. Furthermore, in some periods, the 16th percentile has been below the lower limit of the tolerance range, which is especially evident in the specifications with smoothed expectations. In all specifications except for CPI and unsmoothed expectations, measures of expectations appear to be biased upwards relative to the trend value. Overall, the results suggest that the inflation trend is systematically below the target.

Figure 2. Estimates of trend inflation by specification



Notes: The figure shows the posterior mean of the inflation trend (π_t^*) resulting from the model estimation for each specification, while the bands show the 16th and 84th percentiles of its posterior distribution. The sample period covers 2007Q3 to 2025Q3 (varying due to filtered expectations and core inflation data availability). For sampling, 30,000 draws were taken from the distribution with a burn-in of 1000. The red and gray dashed lines highlights the target value set by the BCCR and the lower limit of the tolerance range, respectively.

Finally, for the third exercise, the linearity hypothesis is rejected for equation (17) at the 10% significance level⁴. This allows for a structural estimation of equation (16). The estimation results are included in Table 2. Importantly, the coefficients on the output gap (d_2, γ) are not statistically different from zero, which is coherent with the strict inflation targeting mandate of the BCCR. Estimates of d_1 and ρ are positive and significant, in line with standard New Keynesian monetary theory. The positive and statistically significant estimate of α ($\alpha \approx 0.28$) implies that the central bank suffers a higher utility loss from inflation values over the target than from values below it. Figure 3 plots the BCCR's loss and reaction functions for the estimated values, against a standard linear policy rule benchmark. As depicted in figure 3a, the loss function has a steeper slope for positive deviations, while it remains relatively flat for negative deviations. The optimal policy rule is therefore more aggressive against positive deviations (see figure 3b). This asymmetry is

⁴The associated Wald statistic yields a value of $W = 5.8444 \sim \chi_2^2$, which implies a p-value of 0.0538. See Appendix B for estimation results.

consistent with the recent critique and effectively creates the contractionary bias, possibly explaining the persistently low inflation trend and expectations observed in the other exercises.

Table 2. Estimates of the Asymmetric Structural Equation

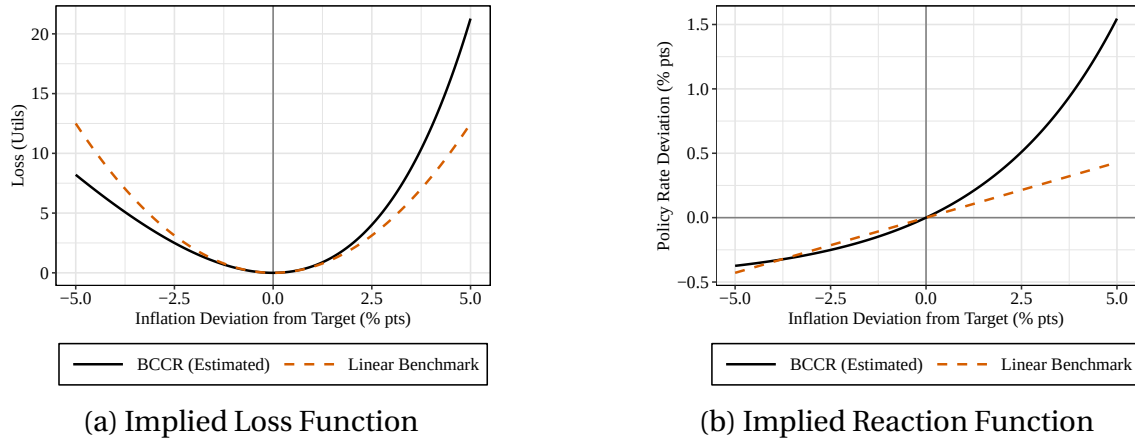
Parameter	Estimate
Constant (d_0)	-0.5971 (0.8517)
Inflation Slope (d_1)	0.1401** (0.0598)
Output Gap Slope (d_2)	0.0002 (0.0037)
Inflation Asymmetry (α)	0.2837*** (0.0151)
Output Gap Asymmetry (γ)	-0.8137 (1.7567)
Interest Smoothing (ρ)	1.0224*** (0.1195)
Observations	77
Sargan–Hansen J -test (p -value)	0.2482

Notes: Standard errors in parentheses. The table reports GMM estimates of the structural Euler equation (16). The inflation target π^T is fixed at 3.0%. Estimation is performed using the Continuously Updated Estimator proposed by Hansen et al. (1996) and a covariance matrix consistent with heteroskedasticity and serial correlation. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

4 Conclusion

This paper presents novel empirical evidence supporting a systematic discrepancy between the BCCR's official inflation target and its actual policy execution. The first exercise reveals that, while long-run expectations are stable, they are anchored to a level significantly below the 3% target. This finding is reinforced by the estimated long-run inflationary trend, which has consistently remained below the target and closer to the lower limit of the tolerance range. The structural evaluation shows evidence of asymmetric preferences that are consistent with the above-mentioned persistent deviations,

Figure 3. Estimated Preferences and Policy Reaction of the BCCR.



Notes: The figure shows the estimated asymmetric loss function and its correspondent reaction function based in equations (15) and (16), respectively, using the estimated values of α and d_1 . The linear benchmark is based on a standard quadratic loss function and an OLS Taylor-rule estimate for the slope.

punishing positive inflation deviations more harshly than negative ones.

The presented evidence aims to support the ongoing discussion on the reevaluation of the inflation target, and suggests that the BCCR has exhibited an implicit contractionary bias relative to its official mandate. This approach has successfully stabilized inflation, but it still poses a risk to the transparency of the monetary policy framework by maintaining a wedge between the public mandate and its execution. While this study maintains a strictly positive focus rather than a normative one, possible alternatives to close this gap include recommitting to a symmetric framework around the 3% target or explicitly lowering the official target to consistently reflect the behavior of the institution.

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Appendices

A VAR Estimation for Long-run Anchor

Table 3. VAR Estimation Results

Regressor	Dependent Variable	
	Inflation (π_t)	Expectations (z_t)
<i>CPI Inflation</i>		
Lagged Inflation	0.4681*** (0.1121)	0.0162*** (0.0054)
Lagged Expectations	0.0369 (1.0199)	0.8202*** (0.0495)
Constant	1.2135 (2.6632)	0.4024*** (0.1292)
Observations	67	67
Adjusted R^2	0.1944	0.4171
<i>Core Inflation</i>		
Lagged Inflation	0.6584*** (0.0966)	0.0417*** (0.0107)
Lagged Expectations	-0.4205 (0.4547)	0.7717*** (0.0504)
Constant	1.7136 (1.1408)	0.4817*** (0.1265)
Observations	67	67
Adjusted R^2	0.4171	0.8358

Notes: Standard errors in parentheses. The table reports the OLS estimates for the reduced-form VAR(1) equations (2, 3). The sample period covers 2009Q2 to 2025Q4. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively.

B Linear Estimation to test for Policy Non-linearity

Table 4. Estimates of the Reduced-Form GMM Equation

Parameter	Estimate
Constant	0.7274 (0.9705)
Inflation	-0.1770** (0.2255)
Output Gap	-0.1355 (0.2392)
Squared Inflation	0.0520** (0.0215)
Squared Output Gap	0.1850** (0.0932)
Interest Smoothing	0.5907** (0.2365)
Observations	77
Sargan–Hansen J -test (p -value)	0.1111
Wald Test (p -value)	0.0538

Notes: Standard errors in parentheses. The table reports GMM estimates of the reduced-form non-linear Taylor rule (17). The sample period covers 2006Q3 to 2025Q3. Estimation is performed using the Continuously Updated Estimator proposed by Hansen et al. (1996) and a covariance matrix consistent with heteroskedasticity and serial correlation. The Wald Test reports the p -value for the joint hypothesis that the quadratic terms are zero ($c_3 = c_4 = 0$). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.